

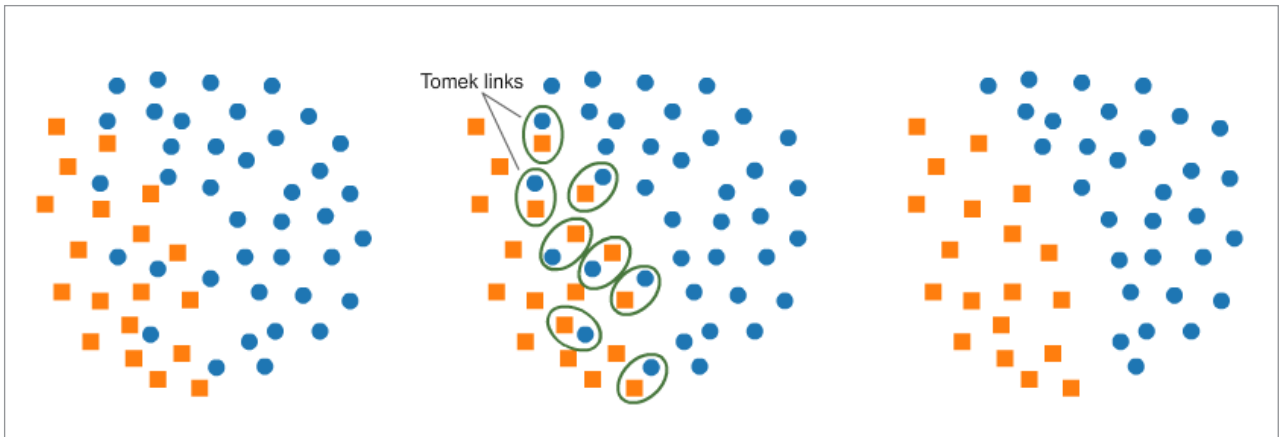


Figure 6: Tomek representation (Image credits: <https://www.analyticsvidhya.com/>)

- The removal of instances involved in Tomek links can enhance the performance of a machine learning model by reducing noise in the dataset and improving the separation between classes.
- The source code is:

```
from collections import Counter
from sklearn.datasets import make_classification
from sklearn.utils import resample
from imblearn.under_sampling import TomekLinks
from matplotlib import pyplot
from numpy import where

X, y = make_classification(n_classes=2, class_sep=2,
weights=[0.1, 0.9], n_features=4, n_clusters_per_class=1,
n_samples=1000, random_state=10)

# summarize class distribution
print('Original dataset shape %s' % Counter(y))
Original dataset shape Counter({1: 894, 0: 106})
```

Next, we will carry out downsampling:

```
cc = TomekLinks()
X_Res, y_res = cc.fit_resample(X, y)
print('Resampled dataset shape %s' % Counter(y_res))
Resampled dataset shape Counter({1: 890, 0: 106})

# scatter plot of examples by class label
for label, _ in Counter(y_res).items():
    row_ix = where(y_res == label)[0]
    pyplot.scatter(X_Res[row_ix, 0], X_Res[row_ix, 1],
label=str(label))
pyplot.legend()
pyplot.show()
```

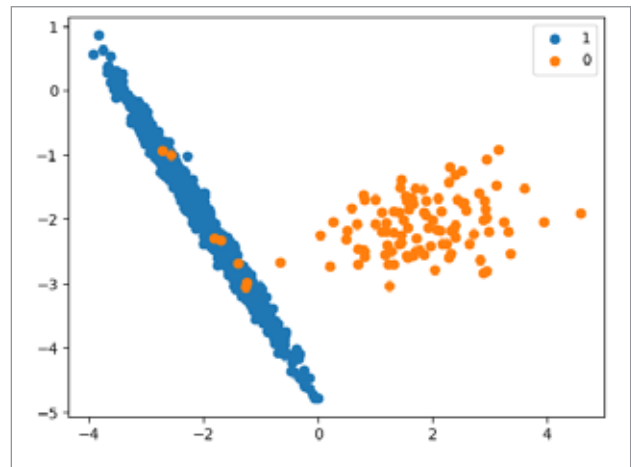


Figure 7: Scatter plot of examples by class label downsampled post using Tomek links

AllKNN (All K-Nearest-Neighbours)

- This method works by removing some of the instances from the majority class that are considered to be 'misclassified' or near the decision boundary.
- The goal is to create a more balanced dataset by eliminating instances from the majority class that may be causing confusion for the classifier.
- The logic is:
 - AllKNN is a modification of RepeatedEditedNearestNeighbours (RENN) that expands the size of the neighbourhood being considered each time it is run.
 - The RENN algorithm runs the EditedNearestNeighbour (ENN) algorithm a specified number of times, removing more outlier points with each rerun.